

Transport Protocols & Frameworks Cheat Sheet

Tech	Conn	Direction	Use	Key Points
HTTP/1.1	TCP	Req→Resp	REST	Text, keep-alive, sequential
HTTP/2	TCP	Req→Resp/Streams	APIs	Binary frames, multiplexing
HTTP/3	QUIC	Req→Resp/Streams	Web	UDP-based, low latency
SSE	HTTP	Server→Client	Notifications	One-way stream
WebSocket	TCP	Bi-dir	Chat	Persistent full-duplex
WebRTC	UDP	Bi-dir	Calls	P2P, media
gRPC	HTTP/2	Unary/Stream	S2S	Protobuf, bidirectional streams
Kafka	TCP	Producer/Consumer	Events	Consumer polls broker
HLS	HTTP	Server→Client	Video	Segment-based CDN delivery

Key points

- HTTP/2 multiplexes many streams over one TCP connection.
- gRPC keeps one HTTP/2 stream open and sends many messages before END_STREAM.
- HTTP/2 transports bytes; gRPC adds message framing.
- SSE is one-way; WebSocket is full-duplex; WebRTC is real-time media.
- HLS uses short video segments over HTTP/CDNs.
- Kafka uses long-poll consumers.